

TWIN COUNTY AP, VA, USA

WMO: 724107

Lat: 36.766N

Lon: 80.823W

Elev: 821

StdP: 91.85

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 53895

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-9.0	-19.9	0.7	-7.4	-17.4	0.9	-6.2	13.3	-2.7	11.9	-4.4	5.4	300	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	30.0	22.2	28.8	21.8	27.6	21.3	24.3	28.0	23.5	26.8	22.7	25.8	2.8	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.6	25.9	22.5	19.0	25.3	22.1	18.5	24.9	78.4	27.9	74.4	26.8	71.6	25.8	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.2	8.0	-15.0	31.9	2.7	1.8	-16.9	33.2	-18.5	34.2	-20.0	35.3	-21.9	36.6		
(5)			-15.9	24.6	2.7	1.2	-17.9	25.5	-19.5	26.3	-21.0	27.0	-23.0	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	1.7	3.1	7.1	12.5	17.0	20.8	22.5	21.9	18.7	13.3	7.4	3.7		
	DBStd	8.67	6.34	5.78	5.68	4.94	3.93	2.84	2.43	2.46	3.33	4.36	5.08	5.43		
	HDD10.0	956	266	200	125	31	2	0	0	0	1	19	108	204		
	HDD18.3	2541	517	426	349	181	73	12	2	3	36	163	327	452		
	CDD10.0	1876	8	8	34	106	218	323	387	370	260	122	31	11		
	CDD18.3	418	0	0	1	6	30	84	131	115	46	7	0	0		
	CDH23.3	2373	0	0	2	37	158	469	788	656	233	29	1	0		
	CDH26.7	489	0	0	0	3	19	92	188	148	37	3	0	0		
(14) Wind	WSAvg	3.0	4.0	4.0	4.0	3.7	2.8	2.3	2.0	1.8	2.1	2.8	3.4	3.6		
(15) Precipitation	PrecAvg	1142	99	71	104	88	104	117	115	106	98	76	79	85		
	PrecMax	1547	213	145	212	152	205	234	193	190	330	240	168	194		
	PrecMin	892	28	26	23	20	32	24	48	33	15	0	12	31		
	PrecStd	169	49	31	43	34	49	56	35	41	65	57	42	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	18.8	22.7	27.1	28.6	31.1	32.4	31.5	29.1	26.3	22.1	18.8		
		MCWB	13.6	14.0	14.8	16.0	20.1	22.5	23.6	23.2	19.9	18.2	15.0	15.0		
	2%	DB	15.1	16.1	19.9	24.0	27.1	28.8	30.1	29.5	27.6	23.7	18.8	16.1		
		MCWB	12.4	11.3	12.9	15.3	19.4	21.5	22.9	22.4	20.2	17.4	13.7	13.1		
	5%	DB	12.8	13.7	17.6	22.4	25.1	27.5	28.7	28.0	26.2	22.1	17.2	13.8		
		MCWB	10.3	9.8	11.9	14.8	18.4	20.9	22.3	21.9	19.8	16.7	13.1	11.3		
	10%	DB	11.0	12.0	15.7	20.4	23.5	26.2	27.5	27.2	24.1	20.0	15.1	12.1		
		MCWB	8.7	8.4	10.8	14.0	17.9	20.2	21.9	21.7	19.3	16.2	11.7	10.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.3	15.4	16.6	18.9	22.2	24.3	25.2	25.5	23.1	21.2	17.8	16.5		
		MCDB	17.1	17.6	20.7	22.2	26.1	28.5	29.1	28.8	26.1	23.6	18.6	17.7		
	2%	WB	13.3	13.0	14.9	17.5	20.8	23.0	24.4	24.2	22.1	19.8	15.7	13.9		
		MCDB	15.0	14.9	18.0	21.5	24.9	26.6	28.4	27.2	25.0	22.1	17.8	15.2		
	5%	WB	11.2	10.8	13.3	16.3	19.9	22.2	23.7	23.2	21.2	18.3	14.1	12.2		
		MCDB	12.4	12.7	16.3	20.3	23.4	25.5	27.3	26.2	24.0	20.1	16.2	13.3		
	10%	WB	8.7	8.7	11.9	15.1	19.0	21.3	22.8	22.5	20.2	17.0	12.4	10.0		
		MCDB	9.8	10.8	14.3	18.9	22.2	24.7	25.8	25.5	23.1	18.7	14.2	11.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.7	9.5	10.1	9.8	9.3	9.1	9.2	9.3	9.6	9.3	8.0		
		MCDBR	9.7	10.9	11.5	11.7	11.3	10.6	10.4	10.5	11.4	11.7	11.3	9.7		
	5% WB	MCWBR	7.3	7.8	7.0	6.1	5.6	5.2	4.5	4.8	5.3	6.2	7.3	7.2		
		MCWBR	8.5	9.6	9.5	9.9	9.5	9.3	9.2	9.2	8.8	9.1	8.4	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.297	0.321	0.359	0.396	0.425	0.456	0.461	0.365	0.332	0.299	0.285		
	taud		2.520	2.482	2.464	2.372	2.272	2.217	2.166	2.128	2.393	2.483	2.541	2.559		
	Ebn at Noon		915	945	948	922	888	858	828	815	887	891	893	897		
	Edh at Noon		78	92	103	120	134	142	149	151	110	91	76	71		
(44) All-Sky Solar Radiation	RadAvg		2.33	2.99	4.07	5.12	5.63	6.09	5.67	5.24	4.42	3.55	2.67	2.01		
	RadStd		0.18	0.36	0.26	0.41	0.46	0.44	0.37	0.27	0.42	0.43	0.24	0.19		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	+141	+86

Nomenclature: See separate page